

# Master Thesis - Outline and Results

Rodrigo Dornelles

## Introduction

- **The lack of legal empirical research:** In Brazil, empirical legal research remains limited. However, it has recently become increasingly common to find studies based on this approach as demonstrated by Cabrera De Bonito et al. (2025).
- **SOTA of legal research:** Those studies, however, still use little data from lawsuits and judicial precedents. Some of them use more qualitative/observational methods. In order to achieve a deeper and broader analysis, it is necessary to have a high level of financial resources and a big and highly qualified research team. Examples of outstanding and important empirical research that showed the relevance of this staff: Associação Brasileira de Jurimetria (2019), Costa et al. (2011), Instituto de Pesquisa Econômica Aplicada (Ipea) & Brasil. Ministério da Justiça e Segurança Pública (2023).
- **Exploratory legal data analysis:** It is possible to analyze the metadata of legal cases. Doing this would be helpful for exploratory analysis and formulating a hypothesis. Data from lawsuits are also used, as some examples show. In Dornelles (2021), data extracted from the Brazilian Supreme Court were used to describe the volume of new actions in the Court during Bolsonaro's administration:

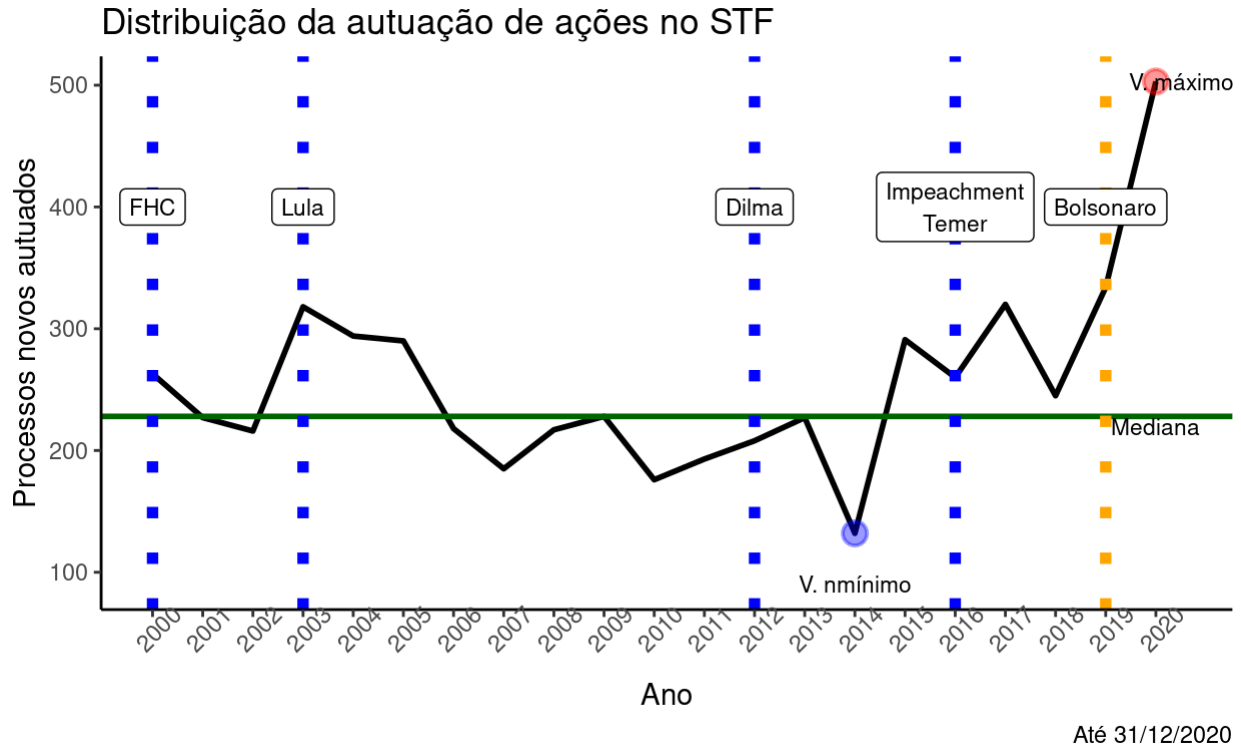


Figure 1: Evolution of new lawsuits each year in the Brazilian Supreme Court. Dornelles (2021)

- **Robust research:** More legal research beyond metadata is needed. To do that, complex documents must be analyzed in depth. Those documents are often relatively large and full of legal jargon, meaning that legal specialists must also do the analysis or at least oversee it.
- **Natural Language Processing:** NLP is an important tool for handling text as data and providing valuable insights from text documents. However, NLP models often specialize in a single task Vajjala et al. (2020), while a handful of tools are necessary to extract information from such documents. Besides that, the traditional NLP models have little ability to understand larger contexts and follow specific instructions Raschka (2024), so large language models would be a better approach to that. (Although there is some debate around Small Language Models and Large Language Models, this distinction is not being made here. See also Chen, 2025).

- **Opportunity:** The advent of powerful language models plus the high level of digitization of Brazil’s Judiciary branch would allow a new era of empirical research, both from the academic and the practitioner’s perspective. It would be possible for a single researcher to study how administrative misconduct law cases were decided in the last few years. Also, it would be possible for a Public Defender to study Local Court decisions to establish a better strategy for new cases.
- **This research:** It aims to understand if and how large language models could be used to leverage legal research by processing judicial decisions and extracting the features of interest. Moreover, the interest is in understanding the feasibility of using open-source large language models in real-world settings.
- **Open-source models:** In academia, in public service, and similar use cases, privacy and costs are limited. Although the terms of use of commercial LLMs suggest a relative degree of confidence about data protection, once inserted in the API, there is no real control over it. Also, commercial models might be expensive (although cheaper than the fair price of human labor to do a similar task), which might block their deployment.<sup>[1]</sup>

## Literature Review

- **Brazilian legal research landscape:** Alves da Silva (2022), Cabrera De Bonito et al. (2025)
- **LLMs applied to legal field:** Junior et al. (2024), Katz et al. (2023), Bertalan & Ruiz (n.d.)
- **LLMs in Portuguese:** Niklaus et al. (2023), Almeida et al. (2024), Pires et al. (2023)
- **Specialization in Legal Tasks:** This is the paper that most gave actionable inputs to this research: Dominguez-Olmedo et al. (2024)

# Research Question

Can open-source large language models be deployed in the Brazilian legal context to accurately extract information, recognize entities, and summarize complex legal documents?

## Data

- **Lacerda Dataset:** Lacerda e Silva (2021) A comprehensive empirical analysis was conducted on over 400 legal cases related to drug trafficking. The researcher collected these cases from the São Paulo State Court of Justice (TJSP), focusing on a sample that covered the years 2017 and 2018. Approximately 35 features were manually annotated and labeled for each case. Access to the dataset was obtained by contacting the researcher, who provided it for further adaptation to meet the specific requirements of this research. Some of the original features also referred to other documents rather than the judicial decision, and, therefore, they were excluded (since it would not be reasonable that the model inferred that). Features related to biological data were also excluded for ethical reasons.

Also, I have tried to access Instituto de Pesquisa Econômica Aplicada (Ipea) & Brasil. Ministério da Justiça e Segurança Pública (2023) micro data to build a golden dataset since they have about 3.5k data points labelled. I have filled a freedom of information request and, although it was possible to access the [general dataset](#), they have not provided the lawsuit identifier, so it was not possible to match each row in their dataset to a legal case and, therefore, to a specific document.

- **Judicial Decisions:** The documents relative to each decision case analyzed by the Lacerda Dataset were extracted automatically from the TJSP website. Since there is no API to that, the process was done using Filho & Trecenti (2020) package called `{tjsp}`. It gathered metadata of each case and the decision in string format in an R

data frame. Some cases were collected by its .pdf files and read automatically.

- **Final dataset:** After cleaning, some duplications were excluded, and some documents were not collected; the final result was a dataset with 258 cases. Also, the final dataset contained 50 columns, which was due to some categorical features being transformed into a boolean. For instance, the column *DenDrogas* contained information about the most common articles of Law in the accusation. It was transformed into the columns *denuncia\_art\_33*, *denuncia\_art\_34*, and *denuncia\_art\_35*.

## Theoretical Framework

- **Natural Language Processing**
- **LLMs**
- **Fine tuning:** Using Dominguez-Olmedo et al. (2024) strategy.
- **LoRA:** Daniel Han & team (2023)

## Models

We used 12 models in this research. OpenAI models were considered commercial, and the others were open source. Also, OpenAI does not disclose its architectures, so it is impossible to know its number of parameters beforehand.

| model        | family | parameters | type       |
|--------------|--------|------------|------------|
| GPT 4.1      | OpenAI | unknown    | commercial |
| GPT 4.1-mini | OpenAI | unknown    | commercial |
| GPT 4.1-nano | OpenAI | unknown    | commercial |
| GPT 4o       | OpenAI | unknown    | commercial |
| GPT 4o-mini  | OpenAI | unknown    | commercial |

|           |        |         |             |
|-----------|--------|---------|-------------|
| o3-mini   | OpenAI | unknown | commercial  |
| Gemma 3   | Gemma  | 27 B    | open-source |
| Gemma 3   | Gemma  | 12 B    | open-source |
| Gemma 3   | Gemma  | 1 B     | open-source |
| Llama 3.1 | Llama  | 8 B     | open-source |
| Llama 3.2 | Llama  | 3 B     | open-source |
| Phi 4     | Phi    | 14 B    | open-source |
| Lawma 8B  | Lawma  | 8 B     | open-source |
| Tucano    | Tucano | 2.4 B   | open-source |

Table 1: Basemodels used

## Methods

- etl
- prompt engeneering, build schema
- generation in OpenAI’s models
- evaluation pipeline
- generation with commercial models
  - window context: rope scaling, yarn, rag, chuncking
  - structured output
  - base model generation
- finetune commercial and open source models
  - OpenAI’s API
  - Unsloth
- final evaluation

# Results

After 17 experiments<sup>1</sup>, the metrics above regarding accuracy were gathered. By accuracy, we are considering the proportion of exact answers relative to the total answers. Below, we can see the performance of the base models before fine-tuning.

| family | type        | model        | parameters | accuracy |
|--------|-------------|--------------|------------|----------|
| Gemma  | open-source | Gemma 3      | 27 B       | 0.8200   |
| OpenAI | commercial  | GPT 4o       | unknown    | 0.7878   |
| OpenAI | commercial  | o3-mini      | unknown    | 0.7638   |
| Gemma  | open-source | Gemma 3      | 12 B       | 0.7627   |
| OpenAI | commercial  | GPT 4.1      | unknown    | 0.7545   |
| Phi    | open-source | Phi 4        | 14 B       | 0.7474   |
| Llama  | open-source | Llama 3.1    | 8 B        | 0.7354   |
| OpenAI | commercial  | GPT 4o-mini  | unknown    | 0.7114   |
| OpenAI | commercial  | GPT 4.1-mini | unknown    | 0.6967   |
| OpenAI | commercial  | GPT 4.1-nano | unknown    | 0.6590   |
| Llama  | open-source | Llama 3.2    | 3 B        | 0.5423   |
| Gemma  | open-source | Gemma 3      | 1 B        | 0.0000   |

Table 2: Accuracy of base models

The model Gemma 3-1B achieved 0.0 accuracy due to its inability to yield valid structured JSON outputs.

The model that best performed in a zero-shot approach was the Gemma 3-27B parameters, which, surprisingly, also beat the benchmark commercial models, even GPT-4o. Also, the Gemma 3 family likely yields accurate results since the lighter version with the 12B parameter

<sup>1</sup>Ignoring validation tests and the experiences to try to overcome the limitations regarding context windows. Also, in this count there were the fine tune.

performed very well and was better than the newly released GPT-4.1. The model Phi-4 also performed very well, betting OpenAI’s smaller models. Regarding the Llama family, the 3.1 version performed better than the 4o-mini and the smaller GPT 4.1-mini and GPT 4.1-nano. On the other hand, Llama 3.2 had a bad performance.

Trying to understand a more fine-grained analysis, breaking the results by the kind of tasks was possible. The *numeric* tasks are those related to extracting and calculating the amount of drugs or prison time in months: those tasks involved not only recognizing the numbers (entities) but also ‘calculating’ the final value. The *boolean* tasks are generally in the realm of question answering in a TRUE/FALSE/None style (for instance, if art. 33 was the basis of the sentence or if there were confessions by the defendant). The *categorical* tasks are those closely related to classification tasks, such as the case’s outcome (conviction, acquittal, etc.). The *open textual* tasks are more likely to be similar to named entity recognition: the name of the Judge and the defendant in this case.

| Accuracy by model and experiment |                  |         |         |             |              |
|----------------------------------|------------------|---------|---------|-------------|--------------|
|                                  | overall_accuracy | numeric | boolean | categorical | open_textual |
| GPT 4o-mini                      | 0.93             | 0.88    | 0.97    | 0.77        | 0.96         |
| Phi 4                            | 0.93             | 0.85    | 0.94    | 0.92        | 0.95         |
| Gemma 3                          | 0.82             | 0.82    | 0.81    | 0.82        | 0.95         |
| GPT 4o                           | 0.79             | 0.88    | 0.75    | 0.87        | 0.96         |
| Llama 3.2                        | 0.78             | 0.74    | 0.78    | 0.81        | 0.85         |
| o3-mini                          | 0.76             | 0.86    | 0.75    | 0.67        | 0.94         |
| Gemma 3                          | 0.76             | 0.74    | 0.77    | 0.66        | 0.94         |
| GPT 4.1                          | 0.75             | 0.88    | 0.71    | 0.86        | 0.96         |
| Phi 4                            | 0.75             | 0.76    | 0.73    | 0.78        | 0.87         |
| Llama 3.1                        | 0.74             | 0.73    | 0.73    | 0.68        | 0.92         |
| GPT 4o-mini                      | 0.71             | 0.77    | 0.67    | 0.81        | 0.95         |
| GPT 4.1-mini                     | 0.70             | 0.83    | 0.64    | 0.82        | 0.91         |
| GPT 4.1-nano                     | 0.66             | 0.80    | 0.61    | 0.68        | 0.96         |
| Llama 3.2                        | 0.54             | 0.46    | 0.58    | 0.33        | 0.64         |

Figure 2: Tasks by accuracy

We can see that the *open textual* tasks have a very high level of performance in nearly all models, while the *boolean* and *categorical* tasks were more diverse performance-wise. We can

also see some differences in numeric tasks, but in general, they all performed very well.

Those results suggested that continuing to work with open-source models was a promising strategy. So, fine-tuning was performed with the models OpenAI 4o-mini (which is the recommended model by OpenAI), Llama 3.2 (3B), and Phi4. No resources were available to perform fine-tuning on other models. After the fine-tuning, we could see a big improvement in the results.

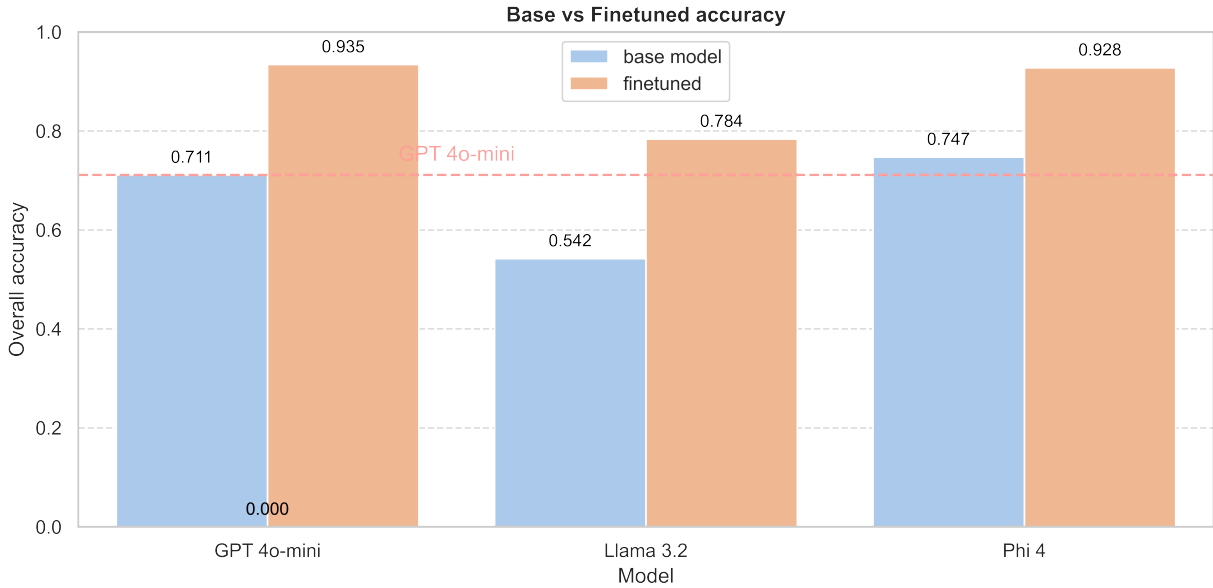


Figure 3: Models after fine-tuning

The process of fine-tuning, even with a small dataset with 186 cases, highly improved the performance of the models, confirming the ‘power of specialization’ as suggested by Dominguez-Olmedo et al. (2024). As Figure 3 showed, model Phi4 tied with GPT 4o-mini and the much smaller Llama 3.2 (3B) learned very easily, even with both models being trained with the same 10 epochs. With a richer dataset and/or more epochs, it would be possible to reach or even surpass GPT 4o-mini performance.

| metric            | GPT 4o-mini | Llama 3.2 | Phi 4 |
|-------------------|-------------|-----------|-------|
| $\Delta$ accuracy | 0.22        | 0.24      | 0.18  |

|                       |       |        |       |
|-----------------------|-------|--------|-------|
| % accuracy            | 31.37 | 44.57  | 24.16 |
| $\Delta$ boolean      | 0.29  | 0.20   | 0.21  |
| % boolean             | 43.56 | 34.54  | 28.63 |
| $\Delta$ categorical  | -0.04 | 0.48   | 0.14  |
| % categorical         | -4.46 | 145.31 | 17.76 |
| $\Delta$ numeric      | 0.11  | 0.28   | 0.09  |
| % numeric             | 14.44 | 62.15  | 11.17 |
| $\Delta$ open textual | 0.01  | 0.21   | 0.08  |
| % open textual        | 1.35  | 33.00  | 8.82  |

Table 3: Fine tuned models: change in accuracy (absolute value  $\Delta$  and relative value %)

As Table 3 shows, Llama 3.2(3B) had the most significant change in its performance, especially with the categorical tasks. Specifically, with this task, we could see a performance degradation of GPT 4o-mini. Phi4, <sup>2</sup> though they had the second-best performance, learned less than Llama, which suggests a ceiling. Assuming that, in those architectures, the number of parameters is the critical factor, fewer parameters than 14B and more than 3B would be ideal.

## Policy Recommendations

- Academia: Disclose data from empirical legal research as much as possible, as it would be a valuable resource for fine-tuning models. If IPEA released the data from Instituto de Pesquisa Econômica Aplicada (Ipea) & Brasil. Ministério da Justiça e Segurança Pública (2023), would it be possible to achieve even better results.
- Open source community: keep at least 128k context window standard and develop light

---

<sup>2</sup>There is the [mini version of Phi-4](#), however it was not possible to run or fine-tune it with the actual computational resources.

open-source models. Wolf et al. (2020) is a fundamental framework for training and inference. Daniel Han & team (2023) was valuable in fine-tuning models efficiently.

- Judiciary branch: should ensure that data and documents from lawsuits are available digitally for research while maintaining high standards of privacy. Anonymization would make this feasible.
- Institutions interested in empirical legal research: OpenAI's API probably offers a better balance of ease and performance. However, open-source models have proven powerful for these tasks and might be deployed to accelerate empirical research in Law.
- Law schools: invest in data literacy for legal students and practitioners.

## References

- Almeida, T. S., Abonizio, H., Nogueira, R., & Pires, R. (2024). *Sabiá-2: A New Generation of Portuguese Large Language Models*. arXiv. <https://doi.org/10.48550/arXiv.2403.09887>
- Alves da Silva, P. E. (2022). Aspectos metodológicos da pesquisa empírica em Direito com processos judiciais físicos e eletrônicos. In *Estudos empíricos em processo e organização judiciária*. Expert.
- Associação Brasileira de Jurimetria. (2019). *Avaliação do Impacto de Critérios Objetivos na Distinção Entre Posse para Uso e Posse para Tráfico: Um estudo Jurimétrico*. Associação Brasileira de Jurimetria. [https://abj.org.br/pdf/20190402\\_abj\\_criterios\\_objetivos.pdf](https://abj.org.br/pdf/20190402_abj_criterios_objetivos.pdf)
- Bertalan, V. G. F., & Ruiz, E. E. S. (n.d.). *Predicting Judicial Outcomes in the Brazilian Legal System Using Textual Features*.
- Cabrera De Bonito, B., Veloso De Castro, C., & Korasaki, V. (2025). PESQUISA QUANTITATIVA NO DIREITO: UMA REVISÃO BIBLIOGRÁFICA. *Revista de Estudos Empíricos Em Direito*, 11. <https://doi.org/10.19092/reed.v11.830>

- Chen, W. (2025). Empowering innovation: The next generation of the Phi family. In *Microsoft Azure Blog*. <https://azure.microsoft.com/en-us/blog/empowering-innovation-the-next-generation-of-the-phi-family/>
- Costa, S. H. da, Silva, P. E. A. da, Sabino, M. A. da C., Fernandes, D. C. M., & Lusvarghi, L. A. dos S. (2011). *A eficácia do sistema jurídico de prevenção e combate à improbidade administrativa* [Publicação de relatório de pesquisa]. Secretaria de Assuntos Legislativos do Ministério da Justiça. [https://web.archive.org/web/20211004195943/http://pensando.mj.gov.br/wp-content/uploads/2015/07/34Pensando\\_Direito1.pdf](https://web.archive.org/web/20211004195943/http://pensando.mj.gov.br/wp-content/uploads/2015/07/34Pensando_Direito1.pdf)
- Daniel Han, M. H., & team, U. (2023). *Unsloth*. <http://github.com/unslothai/unsloth>
- Dominguez-Olmedo, R., Nanda, V., Abebe, R., Bechtold, S., Engel, C., Frankenreiter, J., Gummadi, K., Hardt, M., & Livermore, M. (2024). *Lawma: The Power of Specialization for Legal Tasks*. arXiv. <http://arxiv.org/abs/2407.16615>
- Dornelles, R. F. (2021). O STF, o controle concentrado e a pandemia. In *Associação Brasileira de Jurimetria*. <https://lab.abj.org.br/posts/2021-04-13-stf-controle-concentrado/>
- Filho, J. de J., & Trecenti, J. (2020). *Coleta e organização de dados do Tribunal de Justiça de São Paulo*. <https://jjesufilho.github.io/tjssp/>
- Instituto de Pesquisa Econômica Aplicada (Ipea), & Brasil. Ministério da Justiça e Segurança Pública. (2023). *Perfil do processado e produção de provas nas ações criminais por tráfico de drogas : Relatório analítico nacional dos Tribunais Estaduais de Justiça Comum*. Instituto de Pesquisa Econômica Aplicada (Ipea). <https://doi.org/10.38116/ri221151>
- Junior, R. M., Pires, R., Romero, R., & Nogueira, R. (2024). *Juru: Legal Brazilian Large Language Model from Reputable Sources*. arXiv. <https://doi.org/10.48550/arXiv.2403.18140>
- Katz, D. M., Hartung, D., Gerlach, L., Jana, A., & II, M. J. B. (2023). *Natural Language Processing in the Legal Domain*. arXiv. <https://doi.org/10.48550/arXiv.2302.12039>
- Lacerda e Silva, M. (2021). *Punindo as diferenças: Gênero, raça e geração no sentenciamento*

de tráfico de drogas na cidade de São Paulo. <http://repositorio.unb.br/handle/10482/41413>

Niklaus, J., Matoshi, V., Sturmer, M., Chalkidis, I., & Ho, D. E. (2023). *MultiLegalPile: A 689GB Multilingual Legal Corpus*. <https://doi.org/10.48550/arXiv.2306.02069>

Pires, R., Abonizio, H., Almeida, T. S., & Nogueira, R. (2023). *Sabiá: Portuguese Large Language Models* (Vol. 14197, pp. 226–240). [https://doi.org/10.1007/978-3-031-45392-2\\_15](https://doi.org/10.1007/978-3-031-45392-2_15)

Raschka, S. (2024). *Build a Large Language Model (from Scratch)*. Manning Publications Co. LLC. <http://ebookcentral.proquest.com/lib/hertie-school/detail.action?docID=31657639>

Vajjala, S., Majumder, B., Gupta, A., & Surana, H. (2020). *Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems*. "O'Reilly Media, Inc."

Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Ma, C., Jernite, Y., Plu, J., Xu, C., Le Scao, T., Gugger, S., Drame, M., Lhoest, Q., & Rush, A. M. (2020). *Transformers: State-of-the-Art Natural Language Processing*. 38–45. <https://www.aclweb.org/anthology/2020.emnlp-demos.6>

## Appendix I - Lacerda's Dataset

|                        |                 |
|------------------------|-----------------|
| Name                   | lacerda_example |
| Number of rows         | 420             |
| Number of columns      | 35              |
| Column type frequency: |                 |
| character              | 35              |

Group variables

None

Table 4: Lacerda’s dataset

Table 4: Data summary

**Variable type: character**

| skim_variable | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|---------------|-----------|---------------|-----|-----|-------|----------|------------|
| Processo      | 0         | 1             | 20  | 26  | 0     | 352      | 0          |
| DtDisp        | 0         | 1             | 5   | 10  | 0     | 210      | 0          |
| Juiz          | 0         | 1             | 13  | 49  | 0     | 79       | 0          |
| Vara          | 0         | 1             | 16  | 17  | 0     | 32       | 0          |
| Nome          | 0         | 1             | 11  | 39  | 0     | 419      | 0          |
| Local         | 0         | 1             | 7   | 19  | 0     | 10       | 0          |
| Sexo          | 0         | 1             | 8   | 9   | 0     | 2        | 0          |
| DN            | 0         | 1             | 5   | 28  | 0     | 411      | 0          |
| Cutis         | 0         | 1             | 5   | 7   | 0     | 5        | 0          |
| Cabelo        | 0         | 1             | 4   | 16  | 0     | 8        | 0          |
| Objetos       | 0         | 1             | 4   | 159 | 0     | 261      | 0          |
| Maconha       | 0         | 1             | 1   | 41  | 0     | 255      | 0          |
| Cocaína       | 0         | 1             | 1   | 38  | 0     | 274      | 0          |
| Crack         | 1         | 1             | 1   | 40  | 0     | 138      | 0          |
| Ecstasy       | 0         | 1             | 1   | 24  | 0     | 12       | 0          |
| LSD           | 0         | 1             | 1   | 18  | 0     | 6        | 0          |
| Outras        | 0         | 1             | 1   | 61  | 0     | 33       | 0          |
| DenDrog       | 0         | 1             | 7   | 25  | 0     | 4        | 0          |

| skim_variable           | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|-------------------------|-----------|---------------|-----|-----|-------|----------|------------|
| DenOutros               | 0         | 1             | 3   | 51  | 0     | 13       | 0          |
| Aspectos                | 0         | 1             | 6   | 122 | 0     | 156      | 0          |
| Sentença                | 0         | 1             | 10  | 23  | 0     | 5        | 0          |
| ResDrogas               | 0         | 1             | 2   | 25  | 0     | 6        | 0          |
| ResOutros               | 0         | 1             | 2   | 45  | 0     | 16       | 0          |
| AvalNegPB               | 0         | 1             | 2   | 50  | 0     | 28       | 0          |
| PenaBase                | 0         | 1             | 2   | 12  | 0     | 18       | 0          |
| Agravantes33            | 0         | 1             | 2   | 12  | 0     | 3        | 0          |
| Atenuantes33            | 0         | 1             | 2   | 21  | 0     | 5        | 0          |
| Aumento33               | 0         | 1             | 2   | 26  | 0     | 8        | 0          |
| §4º                     | 0         | 1             | 2   | 13  | 0     | 5        | 0          |
| Pena33                  | 0         | 1             | 2   | 10  | 0     | 46       | 0          |
| PenaDrogas              | 0         | 1             | 2   | 5   | 0     | 7        | 0          |
| PenaOutros              | 0         | 1             | 2   | 9   | 0     | 15       | 0          |
| TotPen                  | 0         | 1             | 2   | 11  | 0     | 59       | 0          |
| Substituição da<br>pena | 0         | 1             | 2   | 3   | 0     | 3        | 0          |
| Regime inicial          | 0         | 1             | 2   | 10  | 0     | 4        | 0          |

Table 5: Lacerda’s dataset

**Appendix II - Prompt**

**Appendix III - Fine tune performance**

**Appendix IV - Plots and tables**

**Appendix V - Acknowledgments**